Blood Bank Management System
Group : 19

Functional Requirements:

User Registration: The system shall allow users to register by providing their required personal information.
User Login: The system shall allow registered users to log in using valid credentials.
Donor Registration: The system shall allow authorized blood-bank staff to register blood donors and record their donor ID, name, blood group, contact information, address, and donation history.
Donor Management: The system shall allow authorized staff to add, update, search, and view donor information.
Blood Donation Recording: The system shall allow blood-bank staff to record each blood donation along with its donation ID, donor ID, blood group, donation date, blood bank, quantity or component, and donation status.
Blood Bank Management: The system shall allow administrators to add, update, and deactivate registered blood banks along with their name, address, location, contact information, and status.
Blood Inventory Management: The system shall allow authorized blood-bank staff to manage the quantity of available blood according to blood group and blood bank.
Inventory Update: The system shall automatically increase or decrease the available blood quantity when blood is added to or removed from the inventory.
Blood Availability Search: The system shall allow users to search for available blood based on blood group, location, and required quantity.
Blood Request: The system shall allow users to submit a blood request by providing the required blood group, quantity, location, urgency, and contact information.
Request Management: The system shall allow authorized blood-bank staff to view and update blood-request statuses such as pending, accepted, rejected, and fulfilled.
Low-Stock Alert: The system shall notify authorized staff when the available quantity of a blood group falls below its configured minimum stock level.
Notifications: The system shall provide notifications for events such as new blood requests, request-status changes, low blood-stock levels, and blood-availability updates.
Reports: The system shall allow authorized users to generate reports about donors, blood donations, blood inventory, blood availability, and blood requests.
Search and Filtering: The system shall allow authorized users to search and filter records using donor ID, donation ID, blood group, blood-bank location, and request ID.
Role-Based Access: The system shall provide different permissions for donors, blood-bank staff, and administrators according to their roles.

Non-Functional Requirements:

Performance: The system shall return blood-availability search results within **2 seconds under normal operating conditions**.
Security: The system shall require authentication for protected functions and shall allow only authorized blood-bank staff to modify blood inventory.
Privacy: The system shall protect the personal information of donors and users from unauthorized access.
Reliability: The system shall maintain accurate blood-inventory information and shall prevent the available blood quantity from becoming negative.
Availability: The system shall be available during the defined operating hours of registered blood banks except during scheduled maintenance.
Usability: The system shall provide a simple and user-friendly interface that allows users to search for blood availability without requiring technical knowledge.
Scalability: The system shall support multiple blood banks, donors, donations, and inventory records without requiring major changes to the system.
Maintainability: The system shall use a modular structure so that individual components can be maintained or modified without affecting the entire system.
Backup and Recovery: The system shall perform regular backups of important donor, donation, and inventory data and shall provide a mechanism for recovering the data after a system failure.
Compatibility: The system shall operate correctly on commonly used modern web browsers such as Chrome, Edge, and Firefox.
Error Handling: The system shall display clear error messages for invalid inputs or unsuccessful operations and shall not crash because of invalid user input.
Auditability: The system shall record important activities such as inventory updates, donation records, and administrative changes along with the user, action, and date/time.
Data Integrity: The system shall maintain consistent and accurate donor, donation, and inventory records and shall prevent duplicate IDs and invalid inventory values.

Functional Requirements SRS table

ID	Requirement	Priority	Actor(s)	Acceptance Criteria
FR-01	The system shall allow users to register by providing required personal information.	High	User/Donor	Valid submission creates an account with a unique user ID.
FR-02	The system shall allow registered users to log in using valid credentials.	High	All Users	Valid credentials grant access; invalid ones are rejected with a clear error.
FR-03	The system shall allow authorized staff to register donors with donor ID, name, blood group, contact, address, and donation history.	High	Blood Bank Staff	Donor record is created with all required fields and a unique donor ID.
FR-04	The system shall allow authorized staff to add, update, search, and view donor records.	High	Blood Bank Staff	All four operations succeed per role permissions.
FR-05	The system shall allow staff to record blood donations with donation ID, donor ID, blood group, date, blood bank, quantity/component, and status.	High	Blood Bank Staff	Donation record is created with all required fields and a unique donation ID.
FR-06	The system shall allow administrators to add, update, and deactivate blood banks with name, address, location, contact, and status.	High	Administrator	Admin can create, modify, and deactivate blood-bank records.
FR-07	The system shall allow authorized staff to manage blood inventory by blood group and blood bank.	High	Blood Bank Staff	Staff can view and update inventory records per role permissions.
FR-08	The system shall automatically update inventory quantity on every donation and issue transaction.	High	System	Inventory reflects the correct quantity after each valid transaction.
FR-09	The system shall allow users to search for available blood by blood group, location, and required quantity.	High	All Users	Search returns matching blood banks within performance target.

FR-10	The system shall allow users to submit blood requests with blood group, quantity, location, urgency, and contact info.	High	User/Requester	Valid request is saved and assigned a unique request ID.
FR-11	The system shall allow authorized staff to view and update blood-request statuses (pending, accepted, rejected, fulfilled).	High	Blood Bank Staff	Staff can view and change status per role permissions.
FR-12	The system shall alert authorized staff when a blood group falls below its configured minimum stock level.	Medium	System/Staff	Alert is generated as soon as inventory crosses threshold.
FR-13	The system shall provide notifications for events including new requests, request-status changes, low stock (FR-12), and availability updates.	Medium	System/All Users	Relevant users receive the correct notification on each configured event.
FR-14	The system shall allow authorized users to generate reports on donors, donations, inventory, availability, and requests.	Medium	Staff/Admin	Authorized users can generate and view the specified reports.
FR-15	The system shall allow authorized users to search and filter records by donor ID, donation ID, blood group, location, and request ID.	Medium	Staff/Admin	Filter returns records matching all selected criteria.
FR-16	The system shall enforce role-based permissions for donors, staff, and administrators.	High	System/Admin	Each role accesses only permitted functions.

Non-Functional Requirements SRS table

ID	Non-Functional Requirement	Priority	Category	Acceptance Criteria

NFR-01	The system shall return blood-availability search results within **2 seconds under normal operating conditions**.	High	Performance	At least the defined normal test workload returns results within 2 seconds.
NFR-02	The system shall require authentication for protected functions and shall allow only authorized blood-bank staff to modify blood inventory.	High	Security	Unauthenticated/unauthorized users cannot access protected inventory functions.
NFR-03	The system shall protect the personal information of donors and users from unauthorized access.	High	Privacy	Unauthorized users cannot access protected personal information.
NFR-04	The system shall maintain accurate blood-inventory information and shall prevent the available blood quantity from becoming negative.	High	Reliability	Inventory never falls below zero after valid or invalid update attempts.
NFR-05	The system shall be available during the defined operating hours of registered blood banks except during scheduled maintenance.	High	Availability	Users can access the system during defined operating hours except during documented maintenance.
NFR-06	The system shall provide a simple and user-friendly interface that allows users to search for blood availability without requiring technical knowledge.	Medium	Usability	A normal user can complete a blood-availability search without technical assistance.

NFR-07	The system shall support multiple blood banks, donors, donations, and inventory records without requiring major changes to the system.	Medium	Scalability	Additional records/blood banks can be added without architectural modification.
NFR-08	The system shall use a modular structure so that individual components can be maintained or modified without affecting the entire system.	Medium	Maintainability	Individual modules can be modified and tested independently where applicable.
NFR-09	The system shall perform regular backups of important donor, donation, and inventory data and shall provide a mechanism for recovering the data after a system failure.	High	Backup & Recovery	Backup can be created and stored, and data can be restored using the recovery mechanism.
NFR-10	The system shall operate correctly on commonly used modern web browsers such as Chrome, Edge, and Firefox.	Medium	Compatibility	Core system functions operate correctly on the specified browsers.
NFR-11	The system shall display clear error messages for invalid inputs or unsuccessful operations and shall not crash because of invalid user input.	High	Error Handling	Invalid input produces an appropriate error message and does not terminate the application.
NFR-12	The system shall record important activities such as inventory updates, donation records, and administrative changes along with the user, action, and date/time.	High	Auditability	Each specified activity creates an audit record containing user, action and timestamp.

NFR-13	The system shall maintain consistent and accurate donor, donation, and inventory records and shall prevent duplicate IDs and invalid inventory values.	High	Data Integrity	Duplicate IDs and invalid inventory values are rejected and records remain consistent.